

Ergonomic Assessment

A nine-sheet observational study of garland-makers and street vendors translating posture, motion, and environment into measurable musculo-skeletal risk.

	A	B
1	WORKSPACE ANALYSIS CHECKLIST	SCORE (1-5)
2		
3	PHYSICAL LAYOUT	
4		
5	-furniture which supports proper posture and movement	1
6	-sufficient space	3
7	-proximity towards the frequently used tools	4
8	-standard ergonomic workspace design	5
9	-safe and easy to use tools	3
10	-adequate amount of light	3
11	-comfortable temperature	2
12	-proper ventilation	5
13	-sufficient storage	5
14	-easy to store	2
15	-efficient and easy accesible to storage	4
16	-easily accesible organisation of material	4
17	-storage that reduces clutter in the optimised space	4
18	-safe to use furniture	1



VERY
POOR



POOR



ACCEPTABLE



GOOD



VERY
GOOD

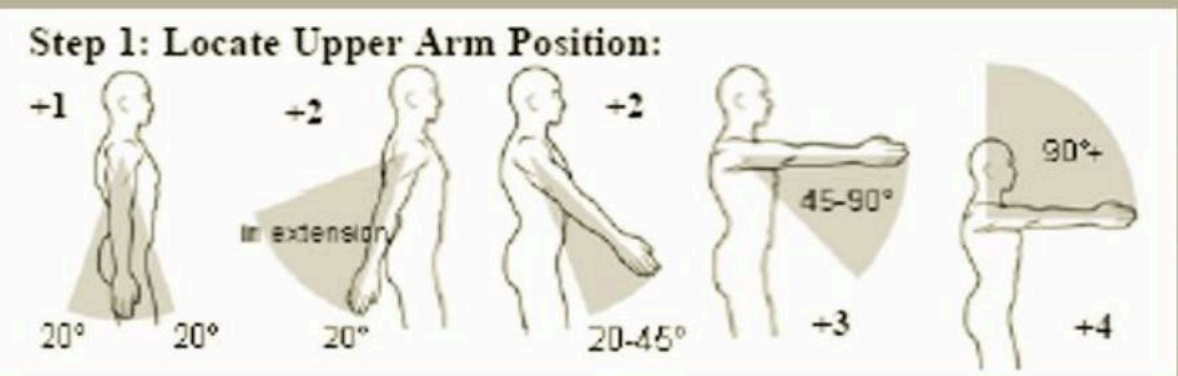
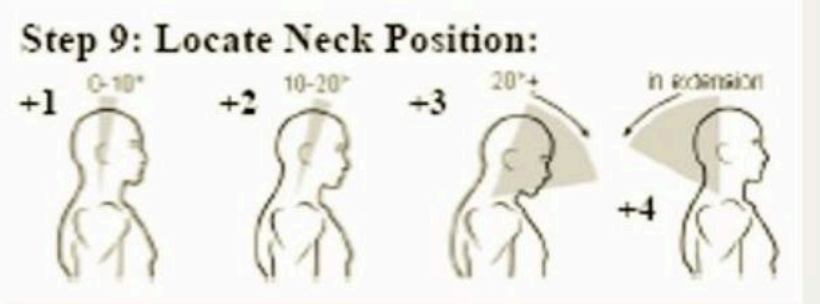
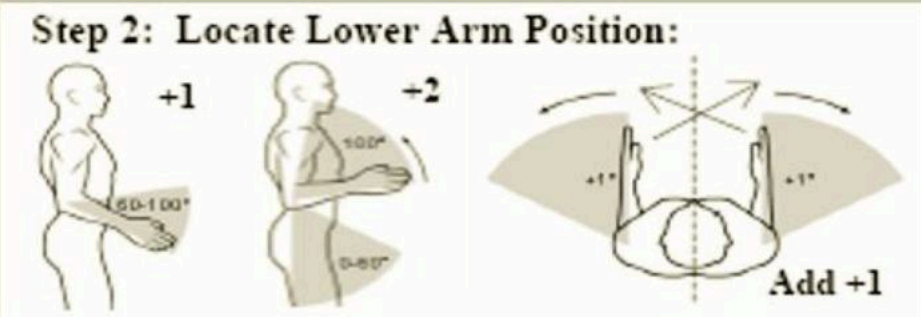
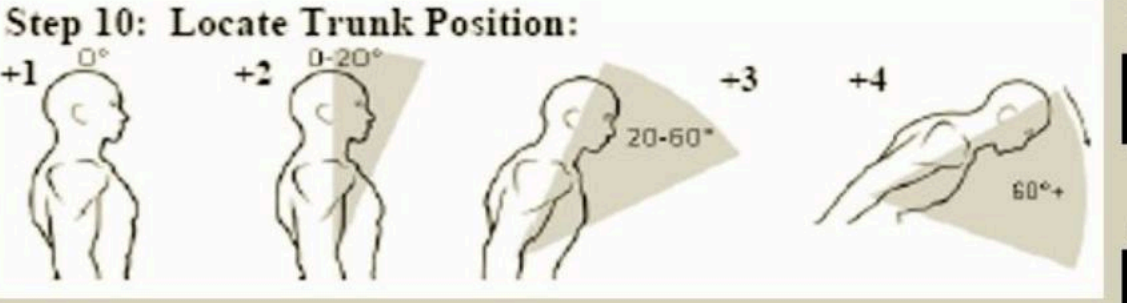
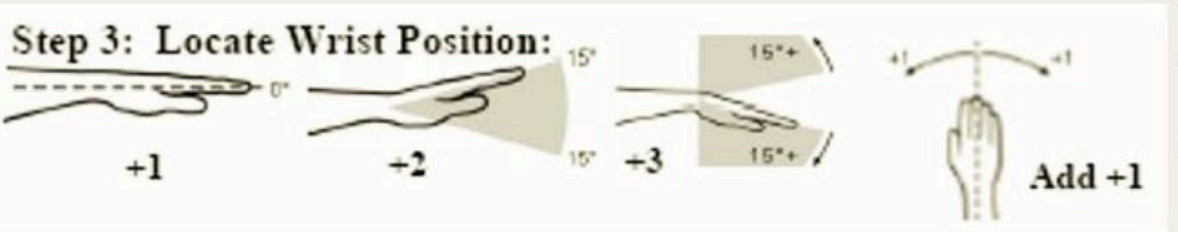
RISK FACTOR ANALYSIS CHECKLIST

Risk Factor	Duration (amount of time a worker is exposed to a risk factor during a shift)			Cause of Risk Factor or Comments	Sub Total
	5 to 59 Minutes	1 to 4 Hours	More than 4 hours		
1. Repetitive motion (every few seconds)	0	1	3	Picking up the flower and putting it into the needle	1
2. Hand force (repetitive force)					
a. Power grip 10+ pounds	1	2	3		-
b. Pinch object	1	2	3	While making the garland	2
3. Awkward postures (repetitive or static)					
Neck: twist/bend	0	1	2	Looking down while making the garland	2
Shoulder: overhead work	0	1	2	While displaying	0
Shoulder: extended reach	0	1	2	While picking flowers, arranging crates, collecting money from customers	2
Elbow/forearm: twist	0	1	2		-
Hand/wrist: bend/twist/pinch	0	1	2	While making the garland	1
Trunk: twist/bend	0	1	2	lifting the crates, cleaning and picking up the flowers	0
Knee: squat/kneel	0	1	2	While cleaning and picking up the flowers	0
4. Using power tools	0	1	2		-
5. Contact stress/hand hammering	0	1	2		-
6. Unsupported fixed postures	1	2	3	While making the garland	2

Risk Factor	Duration (amount of time a worker is exposed to a risk factor during a shift)			Cause of Risk Factor or Comments	Sub Total
	5 to 59 Minutes	1 to 4 Hours	More than 4 hours		
7. Environment (cold temperature, poor lighting, light, and glare)	0	1	2		-
8. No employee control over the work pace	0	1	2		
9. High visibility demands	0	1	2	Due to lack of proper lightning	2
10.Push/pull Easy Moderate Heavy Lift/Lower Carry 10+ Feet	0	1	1		
	0	1	2		0
	1	2	3		
	0	to	6	While lifting the crates filled with flowers	2
	0	to	6	While transporting the garlands from home to their workplace	3
					17

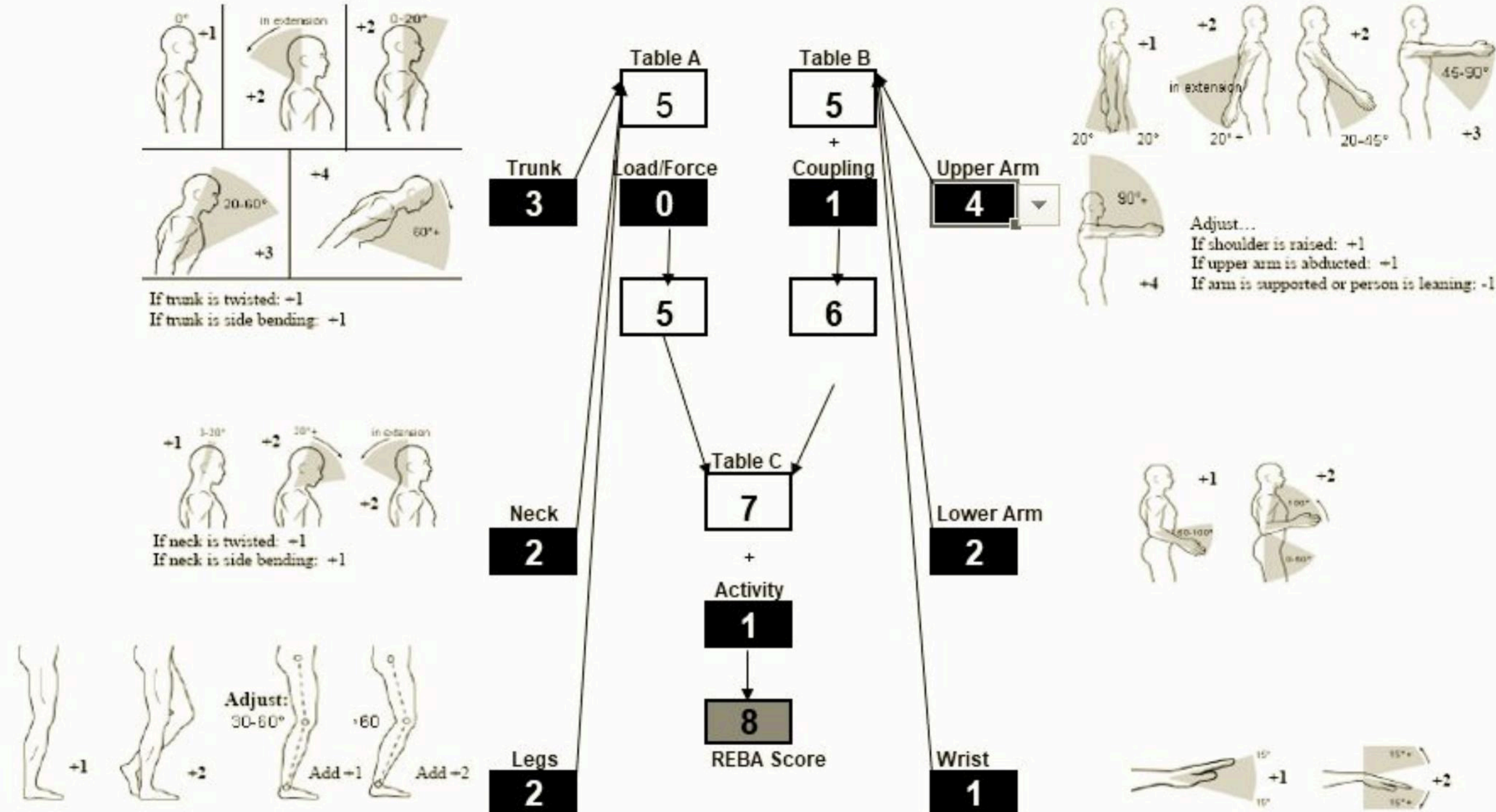
<u>Task</u>	<u>Material Handling</u>	<u>MSDS</u>	<u>Solution</u>
1. Carrying the crates strains the woman vendor's posture	Lifting and transporting the crates	Tunnel syndrome,Tendinitis, Epicondylitis,Muscle StraCarpalins, Tenosynovitis	
2. Garland making causes discomfort to their hands	Stitching garlands	Repetitive Strain Injuries, Tenosynovitis	
3. Furniture available to them is unsafe and uncomfortable to use	Utilizing the furniture	Herniated disc, Knee Bursitis, Sciatica, Hamstring Strain	
4. Varied temperatures makes them feel uneasy	—	—	
5. Process of storing the garlands is taxing for their body	Storing garlands	Back pain, Repetitive Strain Injuries, Hamstring Strain	
6. Garland making posture causes strain to their upper body	Making garlands	Repetitive Strain Injuries, Tenosynovitis	
7. Continuous selling process tenses out their torso	Selling process	Upper Limb disorder, Lower Limb disorder	
8. Segregation of the flowers and the cleaning process is taxing to their body	Segregating the flowers	Rotator cuff syndrome, Trigger finger,Tarsal tunnel syndrome, Sciatica, Epicondylitis, Tendonitis	
9. Standing for prolonged duration causes strain on the calves	Working	Repetitive Strain Injuries, Lower Limb disorder, Back pain	
10. Lack of proper lighting causes strain to their eyes	Working	Eye Strain syndrome	

RULA

Arm & Wrist Analysis		Neck, Trunk & Leg Analysis	
Step 1: Locate Upper Arm Position: 	Upper Arm 2 Upper Arm Adjustment 2	Step 9: Locate Neck Position: 	Neck 3 Neck Adjustment 0
Step 2: Locate Lower Arm Position: 	Lower Arm 2 Lower Arm Adjustment 1	Step 10: Locate Trunk Position: 	Trunk 3 Trunk Adjustment 1
Step 3: Locate Wrist Position: 	Wrist 2	Wrist Twist 1	Legs 1
Arm Muscle Use 1	Force Load Score A 0	Upper Body Muscle Use 1	Force Load Score B 0
Final Score 6			

A. Neck, Trunk and Leg Analysis

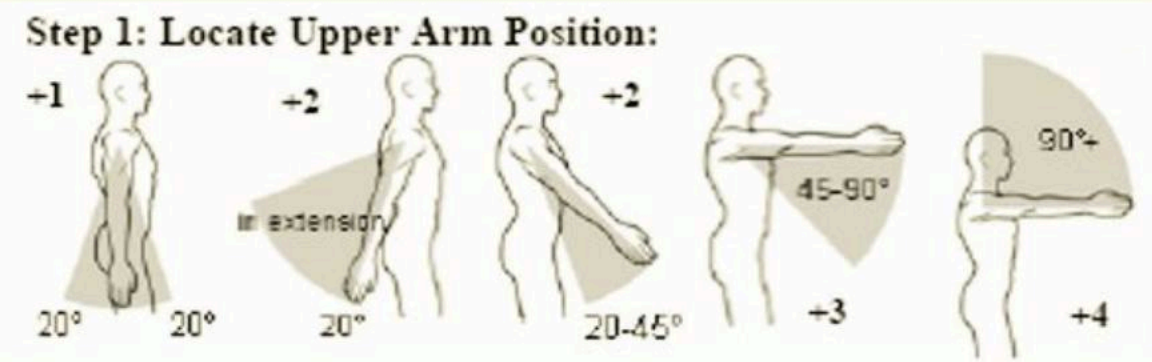
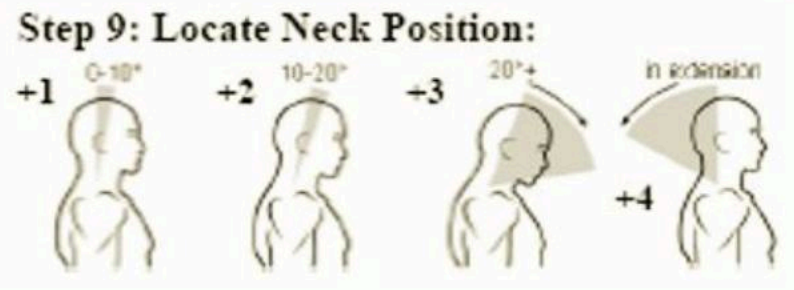
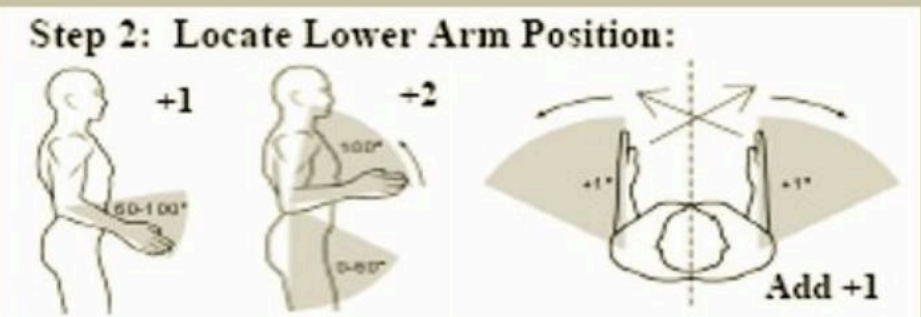
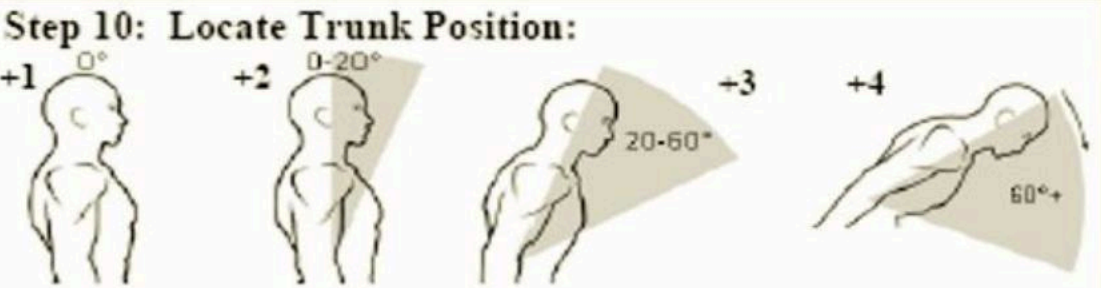
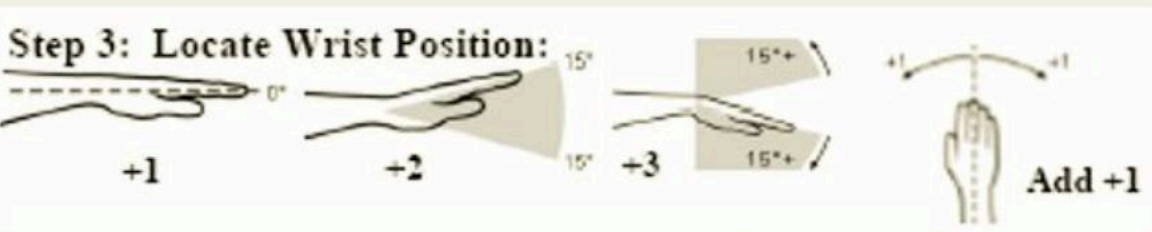
REBA Assessment Worksheet



REBA Score	Risk Level	Action
1	Negligable	None necessary
2 - 3	Low	May be necessary
4 - 7	Medium	Necessary
8 - 10	High	Necessary soon
11 - 15	Very High	Necessary now

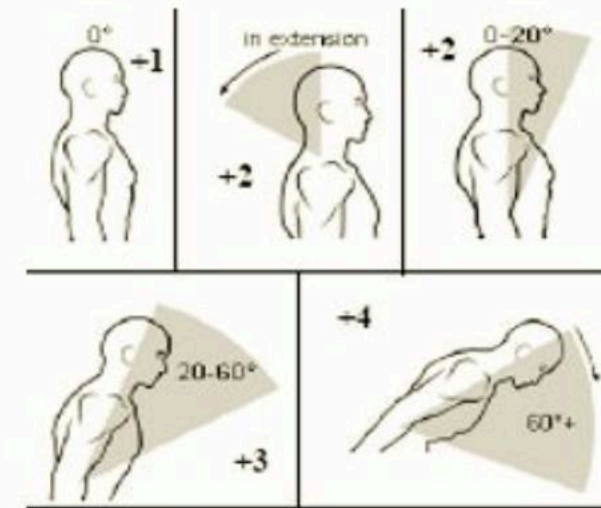
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 Task: XXXXXXXXXXXXXXXXXXXX
 Scorer: XXXXX XXXXX
 Date: dd/mm/yy

RULA

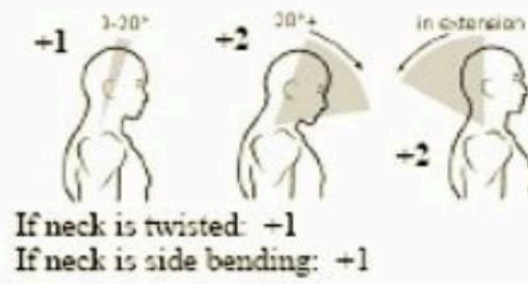
Arm & Wrist Analysis		Neck, Trunk & Leg Analysis	
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Step 2: Locate Lower Arm Position:  Lower Arm 2 Lower Arm Adjustment 1	Step 10: Locate Trunk Position:  Trunk 1 Trunk Adjustment 1		
Step 3: Locate Wrist Position:  Wrist 2	Wrist Twist 1	Legs 2	
Arm Muscle Use 1 Force Load Score A 2	Upper Body Muscle Use 1 Force Load Score B 3		
Final Score		6	

A. Neck, Trunk and Leg Analysis

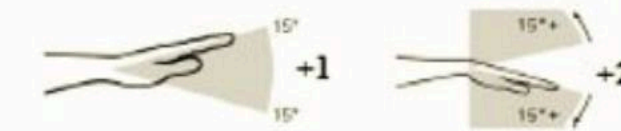
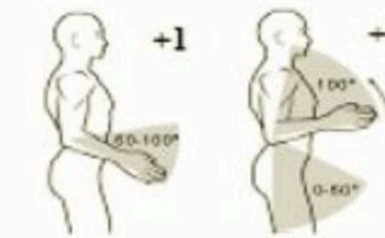
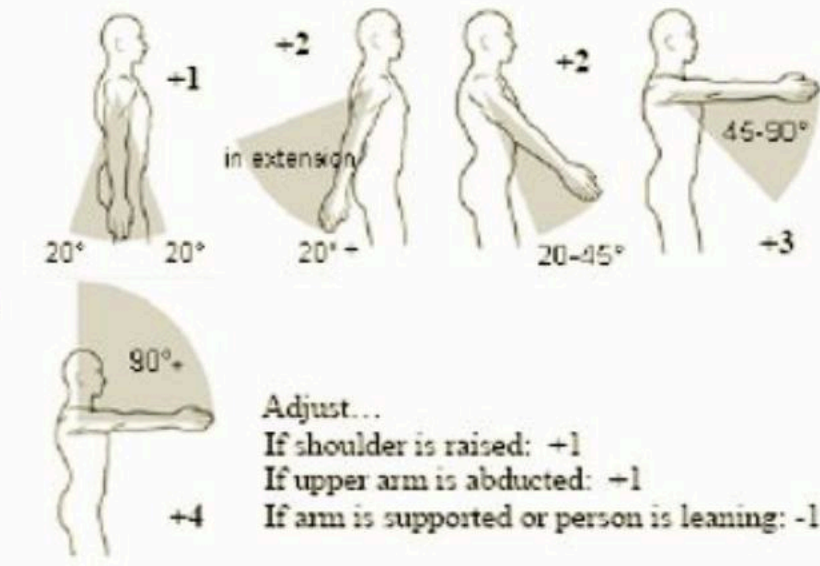
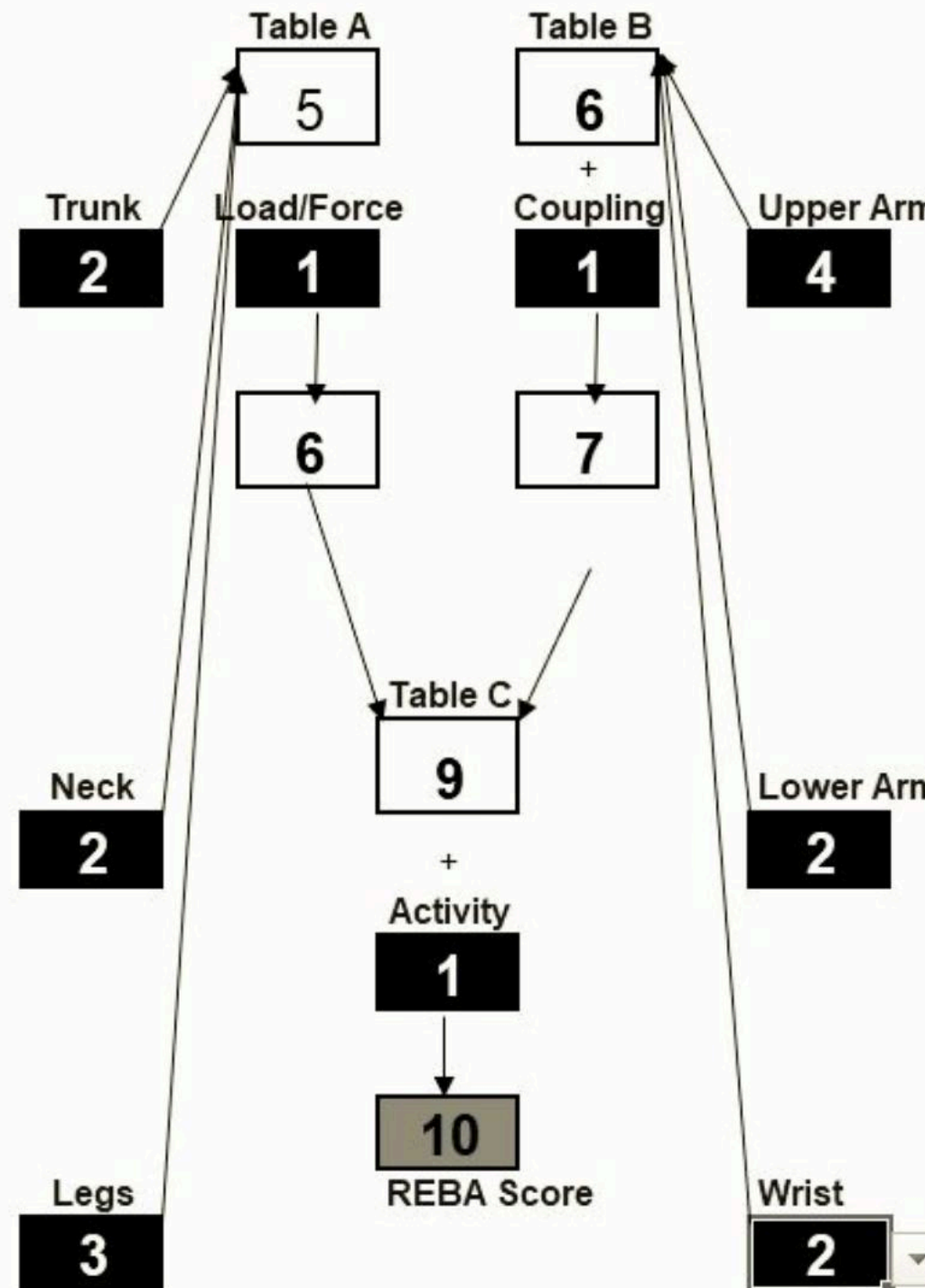
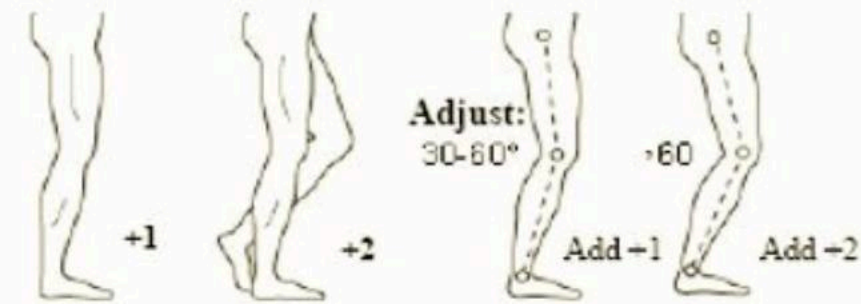
REBA Assessment Worksheet



If trunk is twisted: -1
If trunk is side bending: -1



If neck is twisted: -1
If neck is side bending: -1



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1	Negligable	None necessary
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4 - 7	Medium	Necessary
8 - 10	High	Necessary soon
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Subject: XXXXXXXX
Task: XXXXXXXXXXXXXXXXXXXX
Scorer: XXXXXX XXXXXX
Date: dd/mm/yy

Ergonomic Assessment Checklist		Date	Activity Assessed	
Risk Rating (circle one) <u>High</u> <u>Medium</u> <u>Low</u> *See Notes on bottom of form to obtain the Rating*	Organization		Point of Contact	
	Personnel Observed			
	BLDG NO/Location		ROOM/AREA	
Ergonomic Assessment Checklist				
Risk Factors			Yes	No
1. Have any shop workers been previously diagnosed with any of the following CTD's: Carpal tunnel, Tendonitis, Tenosynovitis, De Quervain's disease, Trigger Finger, White finger, Hand Arm Segmental Vibration Syndrome, Muscle strains, or Back ailments?			☒	☐
2. Have there been any worker complaints concerning ergonomic issues?			☒	☐
3. Do employees perform high repetition tasks? (100 reps/hour to 2000 per/day)			☒	☐
4. Do the employee's routine tasks require repeated heavy lifting? (>20 lbs) or occasional heavy lifting (>50 lbs)			☒	☐
5. Are employees using awkwardly designed tools, which cause the worker to operate the tool outside of a neutral position for an extended period of time? (> 1 hour)			☒	☐
6. Do employees perform tasks with an awkward head or neck position for an extended period of time? (1 to 3 hours)			-	☐
7. Do employees perform tasks that require awkward back angles to be held for extended periods of time (2 to 3 hours)? i.e...hunching, bending, or squatting			-	☐
8. Do employees perform tasks with an awkward elbow angle for an extended period of time (1 to 3 hours) or with extreme force application?			☒	☐
9. Do employees perform tasks with an awkward elbow abduction angle for an extended period of time (1 to 3 hours) or with extreme force application?			☒	☐
10. Do employees perform tasks with an awkward wrist flexion angle for an extended period of time (1 to 3 hours) or with extreme force application?			☒	☐
11. Do employees perform tasks with an awkward wrist extension angle for an extended period of time (1 to 3 hours) or with extreme force application?			☒	☐
12. Do employees perform tasks with an awkward back/hip flexion angle for an extended period of time (1 to 3 hours) or with extreme force application?			-	☐
13. Do employees perform tasks with an extreme reaching distance for an extended period of time (1 to 3 hours) or with extreme force application?			-	☐
14. Do employees perform tasks with an odd work station height (either standing or sitting) for an extended period of time (1-3 hours) or with extreme force application?			☒	☐
15. Are high impact tools used routinely? i.e., riveters, bucking bars, or impact wrenches			-	☐
16. Are high vibration producing tools used routinely? i.e., die grinders, sanders, weed eaters			-	☐
17. Do employees perform tasks at an extreme height (high or low) for an extended period of time (1 to 3 hours) or with extreme force application?			☐	☐
18. Are there any other areas of concern either from your observations or employee complaints?			☒	☐